1 APPLIANCE CONNECTOR / PROTECTED INPUTS

2 AP63205 BUCK -> D11 ISOLATION -> +5V RAIL

3 XIAO 3V3 OUTPUT / CARRIER LOGIC RAIL

4 XIAO ESP32-C3 MODULE / UART / LEVEL SHIFTERS

5 INPUT TEST POINTS

6 RECOVERY HEADER J2 (5V IN VIA D19) / DEBUG HEADER J3

7 PROTECTED-INPUT TVS CLAMP

XIAO ESP32-C3 (22-pad OPL symbol). Native USB stays on the module.
Pad 14 VUSB is tied directly to +5V; pad 21 VBAT is unused.

POWER AND OPERATING NOTES
Appliance pin 1 / pin 3 selection is automatic (U9/U10 load switches, D14/D15 blocking); D1/D2/D3 are input ESD clamps.
U8 AP63205 buck output BUCK_5V reaches the +5V rail through D11, which blocks backfeed into the buck.
XIAO pad 14 VUSB is the module native USB VBUS net and sits directly on +5V; the only fuse is inside the module, downstream of pad 14. XIAO pad 21 VBAT is unused.
J2 is a permanent recovery header: regulated 5V in through D19, 3.3V logic. Use it only with USB and the appliance physically disconnected.
CONNECT ONLY ONE SUPPLY AT A TIME. With the appliance powered the USB connector VBUS is energised; simultaneous sources are not protected.
U9/U10 TPS1H200: FAULT NC, DIAG_EN and exposed pad GND; DELAY via 100k to own VS. CL 3.3k = provisional 0.606 A. Retry 35-45 ms on / 0.8-1.2 s off; bench qualification required. DNP = do not populate; test points and mounting holes are not assembled components.

UNQUALIFIED PROTOTYPE - REVIEW BEFORE FABRICATION
Dr-Crow
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